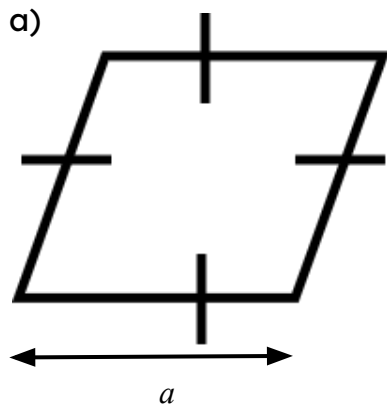




Calculating missing lengths with perimeter of polygons

Task A: Calculating unknown lengths

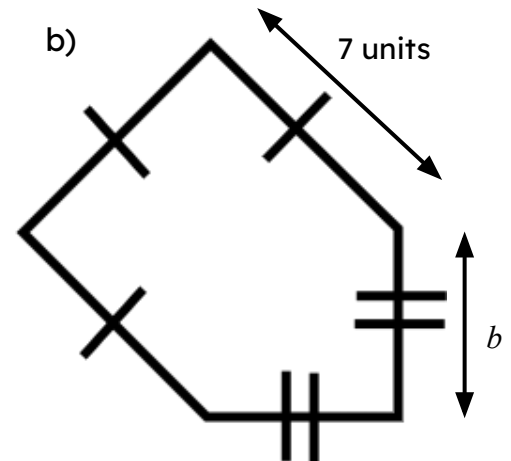
1) Calculate the lengths of the marked sides.



Perimeter = 10 units

$$\begin{aligned} a + a + a + a &= 10 \\ 4a &= 10 \\ a &= \frac{10}{4} \\ a &= 2.5 \end{aligned}$$

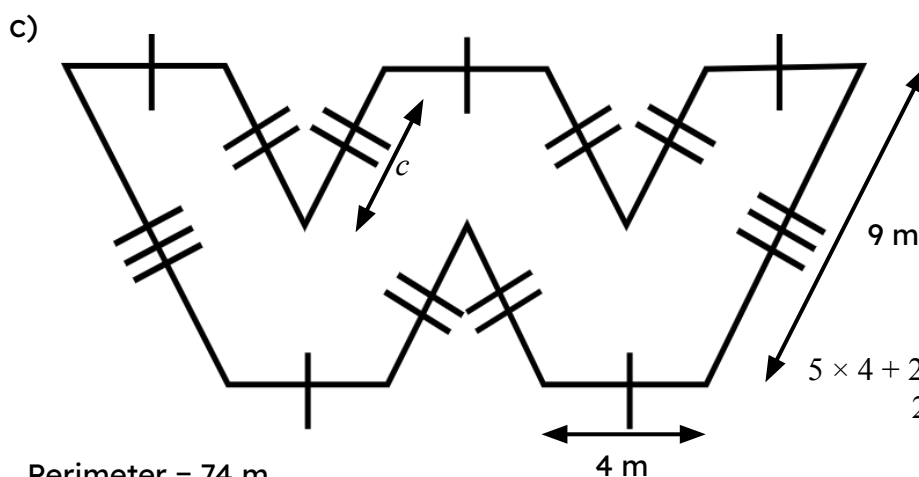
The length of the marked side is 2.5 units.



Perimeter = 25 units

$$\begin{aligned} 7 + 7 + 7 + b + b &= 25 \\ 21 + 2b &= 25 \\ 2b &= 4 \\ b &= 2 \end{aligned}$$

The length of the marked side is 2 units.



Perimeter = 74 m

$$\begin{aligned} 5 \times 4 + 2 \times 9 + 6 \times c &= 74 \\ 20 + 18 + 6c &= 74 \\ 38 + 6c &= 74 \\ 6c &= 36 \\ c &= 6 \end{aligned}$$

The length of the marked side is 6 m.

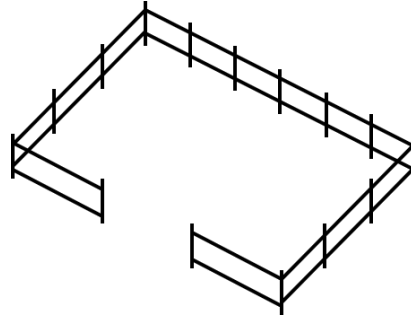
Task B: Problems in context

1) Write your own problem using the same measures but in a different context.

Length = 90 m

Width = 45 m

Perimeter = 195 m



Answers will vary here but should all involve a perimeter of 270 m, a total known length of 195 m and an unknown length which equates to 75 m.